

TEAM ID	SWTID-2026-4030
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	5 MARKS

4.3 – Solution Architecture

The solution architecture describes how electricity consumption data flows through the system, from collection to analysis, forecasting, and visualization.

Architecture Components

1. Data Sources Layer

- Smart Meters
- IoT Sensors
- Utility Company Records
- Historical Electricity Consumption Datasets
- Weather and Environmental Data

2. Data Collection Layer

- Collects real-time and historical electricity usage data.
- Integrates data from multiple sources using APIs and data import tools.

3. Data Storage Layer

- Centralized database for storing electricity consumption records.
- Technologies: MySQL, PostgreSQL, or MongoDB.

4. Data Processing Layer

- Data cleaning and validation.
- Handling missing values and duplicate records.
- Data transformation and normalization.

5. Analytics & Machine Learning Layer

- Consumption pattern analysis.
- Peak load identification.
- Demand forecasting using machine learning algorithms.
- Trend and anomaly detection.

6. Visualization Layer

- Interactive dashboards.
- Graphs, charts, and reports.
- Real-time monitoring interface.

7. User Layer

- Consumers.
- Utility Companies.
- Energy Analysts.
- Government and Regulatory Authorities.

